



Indian Institute of Information Technology, Surat
Department of Computer Science and Engineering
Mid Semester Examination, 3rd Sem, 2025-26
Subject: Operating Systems (CS301)

Timing: 1:30 PM to 3:00 PM

Date: 22/09/2025

Max Marks: 30

Note: Attempt all parts of a question at one place.

1. (a) What is the purpose of System Calls and System Programs? (4)
- (b) Suppose a stack is to be used by the processor to manage procedure calls and returns. Can the program counter be eliminated by using the top of the stack as a program counter? (3)
- (c) Contrast the scheduling policies you might use when trying to optimize a time-sharing system with those you would use to optimize a multiprogrammed batch system. (4)
2. (a) Draw the process state transition diagram and explain each state with an example. (4)
- (b) Explain the concept of context switching in process management. Why is context switching considered overhead, and how does it affect CPU performance? (4)
- (c) Differentiate between a process and a thread, highlighting at least two key differences. Give a suitable example to illustrate your answer. Additionally, explain why threads are considered lightweight compared to processes. (3)
3. Consider the following set of processes with their CPU burst times (in milliseconds) and priorities: (8)

Process	Arrival Time	Burst Time	Priority
P1	0	2	2
P2	1	1	1
P3	2	8	4
P4	3	4	2
P5	4	5	3

- (a) Using Shortest Job First Preemptive (SRTF) scheduling, construct the Gantt chart and calculate the Waiting Time (WT) and Turnaround Time (TAT) for each process. Also, compute the average Waiting Time and average Turn Around Time. (3)
- (b) Using Round Robin (RR) scheduling with quantum = 3 ms, construct the Gantt chart and calculate the Waiting Time (WT) and Turnaround Time (TAT) for each process, along with their averages. (3)
- (c) Compare and analyze the performance of SRTF and RR scheduling. Which algorithm gives better results in terms of average waiting time and turnaround time? Briefly justify your answer. (2)